WORKED SOLUTION

STEP I 1988, Question 16

Probability and Statistics • Continuous Distributions and Conditioning

STEP I 1988, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q16

The key idea

The first part is a normalisation-and-integration exercise: the density is only given up to a constant, so we must find that constant by insisting the total probability is 1, and the integral of $t^2/(1+t^2)^2$ yields to splitting it into $1/(1+t^2)$ and the derivative of $t/(1+t^2)$. After that the testing scheme is a short tree of cases. Only the tests at ages 4 and 5 can matter (no car is rusty before age 3), so we split according to whether the car was already rusty at its fourth test, and the last part is Bayes' theorem on the same tree.

Solution

The distribution of the onset of rust

Let the car become rusty at age $3+T$, where $T \geq 0$ has density $f(t) = k\,t^2/(1+t^2)^2$ for some constant k. The key integral comes from noticing that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{t}{1+t^2}\right) = \frac{1-t^2}{(1+t^2)^2}, \qquad \text{so} \qquad \frac{t^2}{(1+t^2)^2} = \frac{1}{2}\left(\frac{1}{1+t^2} - \frac{1-t^2}{(1+t^2)^2}\right),$$

which you can confirm by putting the bracket over the common denominator $(1+t^2)^2$: the numerator is $(1+t^2)-(1-t^2) = 2t^2$. Hence

$$\int_0^t \frac{s^2}{(1+s^2)^2}\,\mathrm{d}s = \frac{1}{2}\left(\arctan t - \frac{t}{1+t^2}\right).$$

As $t \to \infty$, $\arctan t \to \frac{\pi}{2}$ and $t/(1+t^2) \to 0$, so the integral over $[0,\infty)$ is $\frac{\pi}{4}$. For f to be a density we need $k \cdot \frac{\pi}{4} = 1$, so $k = 4/\pi$. Therefore

$$\boxed{F(t) = \mathrm{P}(\text{rusty before age } 3+t) = \mathrm{P}(T < t) = \frac{2}{\pi}\left(\arctan t - \frac{t}{1+t^2}\right), \quad t \geq 0.}$$

(Notice $F(t) \to 1$: every car rusts eventually.)

Two values we need are

$$F(1) = \frac{2}{\pi}\left(\frac{\pi}{4} - \frac{1}{2}\right) = \frac{1}{2} - \frac{1}{\pi} \approx 0.1817,$$
$$F(2) = \frac{2}{\pi}\left(\arctan 2 - \frac{2}{5}\right) \approx 0.4502.$$

Failing the fifth-anniversary test

Before age 3 no car is rusty, so the tests at ages 1, 2 and 3 are passed for certain (the chance of rust starting at the exact moment of the third test is zero). To fail at age 5 a car must pass at age 4 and fail at age 5, and failing requires being rusty at age 5. Rust never goes away, so there are two disjoint ways this can happen.

- *Rusty already at age* 4 (probability $F(1)$). It passes the fourth test with probability $\frac{1}{2}$ and then, still rusty, fails the fifth with probability $\frac{1}{2}$. Contribution: $\frac{1}{4}F(1)$.
- *Rust begins between ages* 4 *and* 5 (probability $F(2) - F(1)$). It passes the fourth test for certain and fails the fifth with probability $\frac{1}{2}$. Contribution: $\frac{1}{2}\big(F(2) - F(1)\big)$.

Here we are assuming, as the question intends, that the outcomes of different tests on a rusty car are independent of each other and of the time rust began.

> **Ask yourself** Why don't we multiply by the probability of passing the tests at ages 1, 2 and 3?

Because those tests happen before any car can be rusty, so every car passes them; that factor is 1.

Adding the two cases,

$$\begin{aligned}\mathrm{P}(\text{fails at age 5}) &= \tfrac{1}{4}F(1) + \tfrac{1}{2}\big(F(2) - F(1)\big) = \tfrac{1}{2}F(2) - \tfrac{1}{4}F(1)\\ &= \frac{1}{\pi}\left(\arctan 2 - \frac{2}{5}\right) - \frac{1}{8} + \frac{1}{4\pi}\\ &= \boxed{\frac{1}{\pi}\left(\arctan 2 - \frac{3}{20}\right) - \frac{1}{8} \approx 0.1797.}\end{aligned}$$

Rusty at the fourth test, given destruction at the fifth

A car destroyed straight after its fifth test is exactly a car that failed that test (having passed all earlier ones), which is the event just found. Let A be the event that it was rusty at its fourth test and D the event that it failed the fifth. The first bullet above is precisely $A \cap D$, so

$$\mathrm{P}(A \cap D) = \tfrac{1}{4}F(1) = \frac{1}{8} - \frac{1}{4\pi} = \frac{\pi - 2}{8\pi}.$$

By the definition of conditional probability,

$$\mathrm{P}(A \mid D) = \frac{\mathrm{P}(A \cap D)}{\mathrm{P}(D)} = \frac{\dfrac{\pi - 2}{8\pi}}{\dfrac{1}{\pi}\left(\arctan 2 - \dfrac{3}{20}\right) - \dfrac{1}{8}}.$$

Multiplying top and bottom by 8π,

$$\boxed{\mathrm{P}(A \mid D) = \frac{\pi - 2}{8\arctan 2 - \frac{6}{5} - \pi} = \frac{5(\pi - 2)}{40\arctan 2 - 6 - 5\pi} \approx 0.253.}$$

So only about a quarter of the cars destroyed at five years had slipped through their fourth test while already rusty; most had rusted during the fifth year.

Checks

- $F(0) = 0$ and $F(t) \to 1$ as $t \to \infty$, as a distribution function must. Simpson's rule on $\frac{4}{\pi}\int_0^2 t^2/(1+t^2)^2 \,\mathrm{d}t$ gives 0.45018, matching $F(2)$.
- A simulation of 400 000 cars (sampling the onset time and running the annual tests) gave 0.1789 for failing at five years and 0.255 for the conditional probability, against the exact 0.1797 and 0.2528.
- Sanity: $\mathrm{P}(D)$ must lie below $\frac{1}{2}F(2) \approx 0.225$ (the chance of being rusty at five and then failing, ignoring earlier removal), and it does.

Common slips

- Forgetting to normalise: the density is only *proportional* to $t^2/(1+t^2)^2$, and missing the factor $4/\pi$ throws off every later answer.
- Treating "rusty at age 5" as a single case with probability $F(2)$ and multiplying by $\frac{1}{2}$. That ignores the cars which were rusty at age 4 and were removed at that test.
- In the last part, conditioning on "rusty at age 5" rather than on "failed at age 5".

Beyond STEP

The function F here is a distribution function, and the ratio $f(t)/(1-F(t))$ is the *hazard rate*, the instantaneous chance of rusting given the car has survived so far; it is the central object of survival analysis and reliability theory. The final part is a small example of filtering: inferring a hidden state (rusty or not) from imperfect observations (the tests). Grimmett and Welsh, *Probability: An Introduction*, covers continuous distributions and conditioning in this spirit.

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